

$$\rightarrow dX_t = \left(\underbrace{u(X_t)}_{L_{\text{wim}}} + \underbrace{\frac{\sigma_t^2}{2} \nabla_{\log P_t(X_t)}}_{\text{Score}} \right) dt + \sigma_t dW$$

$$\equiv \frac{dX_t}{dt} = u(X_t)$$

$\nabla_x \log P_t(x|z) \rightarrow$ could be as a target for the score function

$$\rightarrow \mathbb{E}_{x,z,t} \left\{ \underbrace{\| S_{\theta}(x) - \nabla_x \log P_t(x|z) \|_2^2}_{d\text{-dim}} \right\}$$

$$= \mathbb{E}_{x,z,t} \left\{ \| S_{\theta}(x) - \nabla_x \log P_t(x) \|_2^2 \right\} + \text{const}$$

$$P_t(x|z) = \mathcal{N}(\alpha_t z, \beta_t I)$$

$$\nabla \log P_t(x|z) = \frac{-(x - \alpha_t z)}{\beta_t^2}$$

$$\mathbb{E} \left\{ \left\| S_{\theta}(x) + \frac{(x - \alpha_t z)}{\beta_t^2} \right\|_2^2 \right\}$$

$$x = \alpha_t z + \beta_t \epsilon \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\mathbb{E} \left\{ \left\| S_{\theta}(x) + \frac{\epsilon}{\beta_t} \right\|_2^2 \right\} = \mathbb{E} \left\{ \frac{1}{\beta_t^2} \left\| \underbrace{\beta_t S_{\theta}(x)}_{-\epsilon_t^{\theta}(x)} + \epsilon \right\|_2^2 \right\}$$

$$= \mathbb{E} \left\{ \frac{1}{\beta_t^2} \left\| \underbrace{\epsilon_t^{\theta}(x)}_{\epsilon_t^{\theta}(x)} - \epsilon \right\|_2^2 \right\}$$

$t \rightarrow 1$

$\beta_t \rightarrow 0$

$$\rightarrow (*) \quad \min_{\theta} \mathbb{E}_{x, \epsilon, t} \left\{ \left\| \epsilon_t^{\theta}(x) - \epsilon \right\|_2^2 \right\}$$

* How to obtain $u(x)$ & $u(x/z)$ given the score functions $\nabla \log p_t(x)$ & $\nabla \log p_t(x/z)$.

(for Gaussian Prob. Path)

$u(x/z)$

$$\frac{d}{dt} X_t = u(X_t | z)$$

$$X_t = \alpha_t z + \beta_t \epsilon$$

$$\alpha_t z + \beta_t \epsilon = u(\underbrace{\alpha_t z + \beta_t \epsilon}_x | z) \quad x \rightarrow \epsilon = \frac{x - \alpha_t z}{\beta_t}$$

$$\alpha_t z + \beta_t \left(\frac{x - \alpha_t z}{\beta_t} \right) = u(x/z)$$

$$\left(\dot{\alpha}_t - \frac{\alpha_t \dot{\beta}_t}{\beta_t} \right) z + \frac{\dot{\beta}_t}{\beta_t} x = u(x/z)$$

$$\nabla_x \log p_t(x/z) = - \frac{x - \alpha_t z}{\beta_t^2}$$

$$u(x/z) = A \underbrace{\nabla_x \log p_t(x/z)}_{- \frac{x - \alpha_t z}{\beta_t^2}} + B = \left(\dot{\alpha}_t - \frac{\alpha_t \dot{\beta}_t}{\beta_t} \right) z + \frac{\dot{\beta}_t}{\beta_t} x$$

$$\frac{A \alpha_t}{\beta_t^2} = \dot{\alpha}_t - \frac{\alpha_t \dot{\beta}_t}{\beta_t} \Rightarrow A = \frac{\dot{\alpha}_t \beta_t^2}{\alpha_t} - \dot{\beta}_t \beta_t$$

$$x \cdot \frac{1}{\beta_t} = - \frac{\dot{\alpha}_t}{\alpha_t} + \frac{\dot{\beta}_t}{\beta_t} + B = \frac{\dot{\beta}_t}{\beta_t} \Rightarrow B = \frac{\dot{\alpha}_t}{\alpha_t}$$

$$\underline{u_t(x/z)} = \left(\frac{\dot{\alpha}_t}{\alpha_t} \beta_t^2 - \dot{\beta}_t \beta_t \right) \underbrace{\nabla \log p_t(x/z)}_{\text{Cond. score}} + \frac{\dot{\alpha}_t}{\alpha_t}$$

$$u(x) = ?$$

$$u(x) = \int u(x|z) \underbrace{\frac{P_t(x|z) P_{data}(z)}{P_t(x)}}_{P_t(z|x)} dz$$

$$u(x) = \int (A_t \nabla_x \log P_t(x|z) + B_t) \frac{P_t(x|z) P_{data}(z)}{P_t(x)} dz$$

$$\int \nabla_x \log P_t(x|z) \cdot \frac{P_t(x|z) P_{data}(z)}{P_t(x)} dz = ?$$

$$\nabla_x \log P_t(x)$$

$$\frac{\nabla_x P_t(x)}{P_t(x)} = \nabla_x \int \frac{P_t(x|z) \cdot P_{data}(z)}{C_t} dz$$

$$= \int \frac{\nabla_x P_t(x|z)}{P_t(x|z)} \cdot \frac{P_t(x|z) P_{data}(z)}{C_t} dz$$

$$= \int \nabla_x \log P_t(x|z) \cdot \frac{P_t(x|z) P_{data}(z)}{P_t(x)} dz$$

$$\nabla_x \log P_t(x) = \int \nabla_x \log P_t(x|z) \cdot \frac{P_t(x|z) \cdot P_{data}(z)}{P_t(x)} dz$$

$$\Rightarrow u_t(x) = A \nabla \log P_t(x) + B$$

$$(1) \quad S_\theta \approx \nabla_x \log P(x|z)$$

$$(2) \quad u_t^\theta = u_t(x)$$

Training could be done based on SDE, but
inference could be done based on ODE.

SDE $\rightarrow s_\theta \rightarrow u \rightarrow$ ODE with u